

Functional Constraints

Flame competition constraint within pocket regions.

Penalty coefficient 0.0100

Maximum allowable heat flux density ratio (max/min) 100.00

The linear burning rate of the first conditional fuel is greater than that of the second.

Penalty coefficient 0.0100

Constraint on maximum kinetic flame heat flux density.

Penalty coefficient 0.0100

Maximum kinetic flame heat flux density in inter-pocket medium 1.00e+09

Maximum kinetic flame heat flux density in skeleton structure 1.00e+08

Maximum kinetic flame heat flux density in pocket region outside skeleton 1.00e+08

Constraint on skeleton thickness to pore diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to pore diameter ratio 3.00

Constraint on skeleton thickness to large oxidizer particle diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to large oxidizer particle diameter ratio 1.00

Parametric Constraints

Constraint on allowable surface temperature range

Minimum surface temperature 600 K

Maximum surface temperature 900 K

Differential Evolution Algorithm

Population size: 384

Mutation force: 0.5

Crossover probability: 0.9

Number of threads: 32

Termination strategy: OrTerminationStrategy

Group Combustion Solver Parameters

Shared Parameters (All Compositions)

Pre-exponential coefficient for gas phase reactions in inter-pocket medium 9025013163.112925

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas phase reactions in inter-pocket medium 177756.81221082766

Parameter bounds [5.000E+004; 2.500E+005]

Pre-exponential factor for gas-phase reactions in pocket region outside skeleton 2955782.273562477

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions in pocket region outside skeleton 50000.00227654699

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in inter-pocket medium 2.4999997539294414

Parameter bounds [0.000E+000; 2.500E+000]

Reaction order for gas-phase chemistry outside skeleton structure 1.3060058329627608

Parameter bounds [0.000E+000; 2.500E+000]

Enthalpy difference between vaporization and binder decomposition 1516000.3553076829

Parameter bounds [-1.000E+007; 1.000E+007]

Diffusion flame height correlation coefficient 0.1118

Parameter bounds [1.000E-003; 1.000E+001]

Power law exponent for burning rate dependency on A (skeleton thickness) 1.0000

Parameter bounds [1.000E+000; 1.000E+000]

Power law exponent for burning rate dependency on B (pore diameter) 2.0000

Parameter bounds [2.000E+000; 2.000E+000]

Radiation temperature correlation coefficient (average layer temperature) 0.9909

Parameter bounds [0.000E+000; 1.000E+000]

Bas_2 (includes Bas_3, Bas_4) - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 999999683.875077

Parameter bounds [1.000E+000; 1.000E+009]

Activation energy for condensed phase reactions 115336.9329489819

Parameter bounds [5.000E+004; 3.000E+005]

Pre-exponential factor for gas-phase reactions within skeleton structure 4540602580119.637

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions within skeleton structure 214349.42094671336

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in skeleton structure 1.1464159929262019

Parameter bounds [0.000E+000; 2.500E+000]

Coefficient A for metal combustion heat flux density in skeleton structure, 4.0719990266396256E-07

Parameter bounds [1.000E-010; 1.000E-003]

Coefficient B for metal combustion heat flux density in skeleton structure, 1.0384836768099168E-12

Parameter bounds [1.000E-012; 1.000E-005]

Bas_1 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 999753482.9378436

Parameter bounds [1.000E+000; 1.000E+009]

Activation energy for condensed phase reactions 89333.14903132689

Parameter bounds [5.000E+004; 3.000E+005]

Pre-exponential factor for gas-phase reactions within skeleton structure 3523173199788.702

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions within skeleton structure 145594.78781013595

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in skeleton structure 1.2701984892696547

Parameter bounds [0.000E+000; 2.500E+000]

Coefficient A for metal combustion heat flux density in skeleton structure, 9.813179063084531E-08

Parameter bounds [1.000E-010; 1.000E-003]

Coefficient B for metal combustion heat flux density in skeleton structure, 2.2468217828348432E-12

Parameter bounds [1.000E-012; 1.000E-005]

Bas_0 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 999896042.0100961

Parameter bounds [1.000E+000; 1.000E+009]

Activation energy for condensed phase reactions 114516.20889892845

Parameter bounds [5.000E+004; 3.000E+005]

Pre-exponential factor for gas-phase reactions within skeleton structure 477108209.0934187

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions within skeleton structure 55170.74104292452

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in skeleton structure 0.02492889975487204

Parameter bounds [0.000E+000; 2.500E+000]

Coefficient A for metal combustion heat flux density in skeleton structure, 5.685359320563564E-07

Parameter bounds [1.000E-010; 1.000E-003]

Coefficient B for metal combustion heat flux density in skeleton structure, 1.0188338161014527E-12

Parameter bounds [1.000E-012; 1.000E-005]

Optimization Results

Objective function value 0.2647

Total penalty from constraint violations 0.0000

Propellant Composition "Bas_2"

At pressure 1 MPa

Experimental burning rate 16.94 mm/s

Calculated burning rate 8.9 mm/s

First Conditional Fuel

Surface temperature 774.39 K

Linear burning rate 11.37 mm/s

Mass decomposition rate 18.86 kg·s⁻¹·m⁻²

Kinetic flame heat flux density to surface 4.21e+07 W/m²

Kinetic flame height 6.97 μm
Average kinetic flame density 1.77 kg/m^3
Average kinetic flame temperature 2,240.69 K
Binder sublimation heat flux density 4.21e+07 W/m^2
Surface heat balance error 0.0016 W/m^2

Secondary Conditional Fuel

Surface temperature 733.43 K
Linear burning rate 4.21 mm/s
Mass decomposition rate 6.99 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 4.86e+06 W/m^2
Kinetic flame height 33.56 μm
Average kinetic flame density 2.56 kg/m^3
Average kinetic flame temperature 1,549.21 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 2e+07 W/m^2
Kinetic flame height 4.28 μm
Average kinetic flame density 3.42 kg/m^3
Average kinetic flame temperature 1,162.21 K
Heat flux density due to metal combustion, 4.68e+06 W/m^2
Effective thermal conductivity 0.4 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity 0.4 $\text{W}/(\text{m}\cdot\text{K})$
Radiative thermal conductivity 9.35e-06 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.35e-04 m
Pore diameter 5.74e-08 m

Diffusion flame heat flux density 3.17e+06 W/m^2
Diffusion flame height 90.46 μm
Heat flux density to surface within skeletal structure 8.87e+06 W/m^2
Heat flux density to surface outside skeletal structure 3.12e+06 W/m^2
Total heat flux density to surface 1.52e+07 W/m^2
Binder sublimation heat flux density 1.52e+07 W/m^2
Surface heat balance error 0.0025 W/m^2

At pressure 4.06 MPa

Experimental burning rate 35.09 mm/s
Calculated burning rate 14.74 mm/s

First Conditional Fuel

Surface temperature 860.13 K
Linear burning rate 66.96 mm/s
Mass decomposition rate 111.09 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface 2.62e+08 W/m^2
Kinetic flame height 1.09 μm
Average kinetic flame density 7.05 kg/m^3
Average kinetic flame temperature 2,283.57 K
Binder sublimation heat flux density 2.62e+08 W/m^2
Surface heat balance error -0.0019 W/m^2

Secondary Conditional Fuel

Surface temperature 749.24 K
Linear burning rate 6.26 mm/s
Mass decomposition rate 10.39 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 2.04e+07 W/m^2
Kinetic flame height 7.91 μm
Average kinetic flame density 10.34 kg/m^3
Average kinetic flame temperature 1,557.12 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.42e+07 W/m^2
Kinetic flame height 5.92 μm

Average kinetic flame density 13.76 kg/m³
 Average kinetic flame temperature 1,170.12 K
 Heat flux density due to metal combustion, 6.89e+06 W/m²
 Effective thermal conductivity 0.4 W/(m·K)
 Conductive thermal conductivity 0.4 W/(m·K)
 Radiative thermal conductivity 4.51e-06 W/(m·K)
 Conductive thermal conductivity balance error -4.092294297874943E-08
 Skeleton layer thickness 9.08e-05 m
 Pore diameter 2.6e-08 m
 Diffusion flame heat flux density 2.12e+06 W/m²
 Diffusion flame height 134.48 μm
 Heat flux density to surface within skeletal structure 6.97e+06 W/m²
 Heat flux density to surface outside skeletal structure 1.37e+07 W/m²
 Total heat flux density to surface 2.28e+07 W/m²
 Binder sublimation heat flux density 2.28e+07 W/m²
 Surface heat balance error 0.0034 W/m²

At pressure 6.5 MPa

Experimental burning rate 44.85 mm/s
 Calculated burning rate 18.86 mm/s

First Conditional Fuel

Surface temperature 893.51 K
 Linear burning rate 121.79 mm/s
 Mass decomposition rate 202.04 kg·s⁻¹·m⁻²
 Kinetic flame heat flux density to surface 4.87e+08 W/m²
 Kinetic flame height 0.58 μm
 Average kinetic flame density 11.22 kg/m³
 Average kinetic flame temperature 2,300.25 K
 Binder sublimation heat flux density 4.87e+08 W/m²
 Surface heat balance error 0.0021 W/m²

Secondary Conditional Fuel

Surface temperature 758.92 K
 Linear burning rate 7.91 mm/s
 Mass decomposition rate 13.13 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 3e+07 W/m²
 Kinetic flame height 5.36 μm
 Average kinetic flame density 16.52 kg/m³
 Average kinetic flame temperature 1,561.96 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.15e+07 W/m²
 Kinetic flame height 7.23 μm
 Average kinetic flame density 21.96 kg/m³
 Average kinetic flame temperature 1,174.96 K
 Heat flux density due to metal combustion, 8.66e+06 W/m²
 Effective thermal conductivity 0.4 W/(m·K)
 Conductive thermal conductivity 0.4 W/(m·K)
 Radiative thermal conductivity 2.93e-06 W/(m·K)
 Conductive thermal conductivity balance error -4.092294297874943E-08
 Skeleton layer thickness 7.18e-05 m
 Pore diameter 1.63e-08 m
 Diffusion flame heat flux density 1.67e+06 W/m²
 Diffusion flame height 170.01 μm
 Heat flux density to surface within skeletal structure 5.48e+06 W/m²
 Heat flux density to surface outside skeletal structure 2.18e+07 W/m²
 Total heat flux density to surface 2.9e+07 W/m²
 Binder sublimation heat flux density 2.9e+07 W/m²
 Surface heat balance error -0.0025 W/m²

Propellant Composition "Bas_3"

At pressure 1 MPa

Experimental burning rate 15.24 mm/s
Calculated burning rate 15.18 mm/s

First Conditional Fuel

Surface temperature 774.39 K
Linear burning rate 11.37 mm/s
Mass decomposition rate $18.86 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $4.21\text{e}+07 \text{ W/m}^2$
Kinetic flame height 6.97 μm
Average kinetic flame density 1.77 kg/m^3
Average kinetic flame temperature 2,240.69 K
Binder sublimation heat flux density $4.21\text{e}+07 \text{ W/m}^2$
Surface heat balance error 0.0016 W/m^2

Secondary Conditional Fuel

Surface temperature 774.38 K
Linear burning rate 11.37 mm/s
Mass decomposition rate $18.86 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $1.82\text{e}+06 \text{ W/m}^2$
Kinetic flame height 87.63 μm
Average kinetic flame density 2.53 kg/m^3
Average kinetic flame temperature 1,569.69 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $9.61\text{e}+07 \text{ W/m}^2$
Kinetic flame height 1.92 μm
Average kinetic flame density 2.34 kg/m^3
Average kinetic flame temperature 1,698.69 K
Heat flux density due to metal combustion, $6.77\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $2.14\text{e}-06 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-5.5152085010057306\text{E}-08$
Skeleton layer thickness $5\text{e}-05 \text{ m}$
Pore diameter $7.88\text{e}-09 \text{ m}$
Diffusion flame heat flux density $8.65\text{e}+06 \text{ W/m}^2$
Diffusion flame height 33.51 μm
Heat flux density to surface within skeletal structure $3.22\text{e}+07 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $1.25\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $4.21\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $4.21\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0037 W/m^2

At pressure 4.06 MPa

Experimental burning rate 37.34 mm/s
Calculated burning rate 23.61 mm/s

First Conditional Fuel

Surface temperature 860.13 K
Linear burning rate 66.96 mm/s
Mass decomposition rate $111.09 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.62\text{e}+08 \text{ W/m}^2$
Kinetic flame height 1.09 μm
Average kinetic flame density 7.05 kg/m^3
Average kinetic flame temperature 2,283.57 K
Binder sublimation heat flux density $2.62\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.0019 W/m^2

Secondary Conditional Fuel

Surface temperature 774.39 K

Linear burning rate 11.37 mm/s

Mass decomposition rate $18.87 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $1.13\text{e}+07 \text{ W/m}^2$

Kinetic flame height 14.08 μm

Average kinetic flame density 10.25 kg/m^3

Average kinetic flame temperature 1,569.7 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $9.95\text{e}+07 \text{ W/m}^2$

Kinetic flame height 1.86 μm

Average kinetic flame density 9.48 kg/m^3

Average kinetic flame temperature 1,698.7 K

Heat flux density due to metal combustion, $6.77\text{e}+06 \text{ W/m}^2$

Effective thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$

Radiative thermal conductivity $2.14\text{e}-06 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity balance error -5.5152085010057306E-08

Skeleton layer thickness $5\text{e}-05 \text{ m}$

Pore diameter $7.88\text{e}-09 \text{ m}$

Diffusion flame heat flux density $8.65\text{e}+06 \text{ W/m}^2$

Diffusion flame height 33.51 μm

Heat flux density to surface within skeletal structure $2.48\text{e}+07 \text{ W/m}^2$

Heat flux density to surface outside skeletal structure $8.66\text{e}+06 \text{ W/m}^2$

Total heat flux density to surface $4.21\text{e}+07 \text{ W/m}^2$

Binder sublimation heat flux density $4.21\text{e}+07 \text{ W/m}^2$

Surface heat balance error $6.04\text{e}-05 \text{ W/m}^2$

At pressure 6.5 MPa

Experimental burning rate 50.5 mm/s

Calculated burning rate 25.04 mm/s

First Conditional Fuel

Surface temperature 893.51 K

Linear burning rate 121.79 mm/s

Mass decomposition rate $202.04 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Kinetic flame heat flux density to surface $4.87\text{e}+08 \text{ W/m}^2$

Kinetic flame height 0.58 μm

Average kinetic flame density 11.22 kg/m^3

Average kinetic flame temperature 2,300.25 K

Binder sublimation heat flux density $4.87\text{e}+08 \text{ W/m}^2$

Surface heat balance error 0.0021 W/m^2

Secondary Conditional Fuel

Surface temperature 774.69 K

Linear burning rate 11.45 mm/s

Mass decomposition rate $19 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $2.08\text{e}+07 \text{ W/m}^2$

Kinetic flame height 7.65 μm

Average kinetic flame density 16.43 kg/m^3

Average kinetic flame temperature 1,569.85 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $1\text{e}+08 \text{ W/m}^2$

Kinetic flame height 1.85 μm

Average kinetic flame density 15.19 kg/m^3

Average kinetic flame temperature 1,698.85 K

Heat flux density due to metal combustion, $6.81\text{e}+06 \text{ W/m}^2$

Effective thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$

Radiative thermal conductivity $2.12\text{e}-06 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity balance error -5.5152085010057306E-08

Skeleton layer thickness 4.96×10^{-5} m
Pore diameter 7.77×10^{-9} m
Diffusion flame heat flux density 8.59×10^6 W/m²
Diffusion flame height 33.74 μ m
Heat flux density to surface within skeletal structure 1.62×10^7 W/m²
Heat flux density to surface outside skeletal structure 1.76×10^7 W/m²
Total heat flux density to surface 4.24×10^7 W/m²
Binder sublimation heat flux density 4.24×10^7 W/m²
Surface heat balance error -0.0021 W/m²

Propellant Composition "Bas_4"

At pressure 1 MPa

Experimental burning rate 9.73 mm/s
Calculated burning rate 9.89 mm/s

First Conditional Fuel

Surface temperature 774.39 K
Linear burning rate 11.37 mm/s
Mass decomposition rate $18.86 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$
Kinetic flame heat flux density to surface 4.21×10^7 W/m²
Kinetic flame height 6.97 μ m
Average kinetic flame density 1.77 kg/m^3
Average kinetic flame temperature 2,240.69 K
Binder sublimation heat flux density 4.21×10^7 W/m²
Surface heat balance error 0.0016 W/m²

Secondary Conditional Fuel

Surface temperature 724.86 K
Linear burning rate 3.37 mm/s
Mass decomposition rate $5.6 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 131,802.77 W/m²
Kinetic flame height 240.62 μ m
Average kinetic flame density 4.49 kg/m^3
Average kinetic flame temperature 883.43 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 5.33×10^6 W/m²
Kinetic flame height 9.68 μ m
Average kinetic flame density 4.04 kg/m^3
Average kinetic flame temperature 982.93 K
Heat flux density due to metal combustion, 1.32×10^7 W/m²
Effective thermal conductivity $1.41 \text{ W/(m} \cdot \text{K)}$
Conductive thermal conductivity $1.41 \text{ W/(m} \cdot \text{K)}$
Radiative thermal conductivity $1.11 \times 10^{-5} \text{ W/(m} \cdot \text{K)}$
Conductive thermal conductivity balance error $9.240271880983641 \times 10^{-10}$
Skeleton layer thickness 1.69×10^{-4} m
Pore diameter 8.95×10^{-8} m
Diffusion flame heat flux density 2.22×10^6 W/m²
Diffusion flame height 128.81 μ m
Heat flux density to surface within skeletal structure 9.79×10^6 W/m²
Heat flux density to surface outside skeletal structure 62,109.73 W/m²
Total heat flux density to surface 1.21×10^7 W/m²
Binder sublimation heat flux density 1.21×10^7 W/m²
Surface heat balance error 2.16×10^{-4} W/m²

At pressure 4.06 MPa

Experimental burning rate 25.92 mm/s
Calculated burning rate 12.1 mm/s

First Conditional Fuel

Surface temperature 860.13 K

Linear burning rate 66.96 mm/s
Mass decomposition rate $111.09 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.62\text{e}+08 \text{ W/m}^2$
Kinetic flame height $1.09 \mu\text{m}$
Average kinetic flame density 7.05 kg/m^3
Average kinetic flame temperature 2,283.57 K
Binder sublimation heat flux density $2.62\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.0019 W/m^2

Secondary Conditional Fuel

Surface temperature 726.1 K
Linear burning rate 3.48 mm/s
Mass decomposition rate $5.78 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $794,245.45 \text{ W/m}^2$
Kinetic flame height $39.77 \mu\text{m}$
Average kinetic flame density 18.21 kg/m^3
Average kinetic flame temperature 884.05 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $5.35\text{e}+06 \text{ W/m}^2$
Kinetic flame height $9.62 \mu\text{m}$
Average kinetic flame density 16.37 kg/m^3
Average kinetic flame temperature 983.55 K
Heat flux density due to metal combustion, $1.36\text{e}+07 \text{ W/m}^2$
Effective thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $1.05\text{e}-05 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $9.240271880983641\text{E}-10$
Skeleton layer thickness $1.63\text{e}-04 \text{ m}$
Pore diameter $8.39\text{e}-08 \text{ m}$

Diffusion flame heat flux density $2.14\text{e}+06 \text{ W/m}^2$
Diffusion flame height $133.05 \mu\text{m}$
Heat flux density to surface within skeletal structure $9.95\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $377,282.72 \text{ W/m}^2$
Total heat flux density to surface $1.25\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $1.25\text{e}+07 \text{ W/m}^2$
Surface heat balance error $-8.9\text{e}-04 \text{ W/m}^2$

At pressure 6.5 MPa

Experimental burning rate 36.06 mm/s
Calculated burning rate 12.63 mm/s

First Conditional Fuel

Surface temperature 893.51 K
Linear burning rate 121.79 mm/s
Mass decomposition rate $202.04 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $4.87\text{e}+08 \text{ W/m}^2$
Kinetic flame height $0.58 \mu\text{m}$
Average kinetic flame density 11.22 kg/m^3
Average kinetic flame temperature 2,300.25 K
Binder sublimation heat flux density $4.87\text{e}+08 \text{ W/m}^2$
Surface heat balance error 0.0021 W/m^2

Secondary Conditional Fuel

Surface temperature 727.1 K
Linear burning rate 3.58 mm/s
Mass decomposition rate $5.93 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $1.43\text{e}+06 \text{ W/m}^2$
Kinetic flame height $21.98 \mu\text{m}$
Average kinetic flame density 29.17 kg/m^3
Average kinetic flame temperature 884.55 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $5.29\text{e}+06 \text{ W/m}^2$
Kinetic flame height $9.72 \text{ }\mu\text{m}$
Average kinetic flame density 26.22 kg/m^3
Average kinetic flame temperature 984.05 K
Heat flux density due to metal combustion, $1.4\text{e}+07 \text{ W/m}^2$
Effective thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $9.97\text{e}-06 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $9.240271880983641\text{E}-10$
Skeleton layer thickness $1.59\text{e}-04 \text{ m}$
Pore diameter $7.97\text{e}-08 \text{ m}$
Diffusion flame heat flux density $2.09\text{e}+06 \text{ W/m}^2$
Diffusion flame height $136.57 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $1\text{e}+07 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $685,187.11 \text{ W/m}^2$
Total heat flux density to surface $1.28\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $1.28\text{e}+07 \text{ W/m}^2$
Surface heat balance error $1.93\text{e}-04 \text{ W/m}^2$

Propellant Composition "Bas_1"

At pressure 1 MPa

Experimental burning rate 14.26 mm/s
Calculated burning rate 18.51 mm/s

First Conditional Fuel

Surface temperature 602.35 K
Linear burning rate 10.8 mm/s
Mass decomposition rate $17.92 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $3.53\text{e}+07 \text{ W/m}^2$
Kinetic flame height $8.78 \text{ }\mu\text{m}$
Average kinetic flame density 1.84 kg/m^3
Average kinetic flame temperature $2,154.68 \text{ K}$
Binder sublimation heat flux density $3.53\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0051 W/m^2

Secondary Conditional Fuel

Surface temperature 602.35 K
Linear burning rate 10.8 mm/s
Mass decomposition rate $17.92 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $1.83\text{e}+06 \text{ W/m}^2$
Kinetic flame height $96.55 \text{ }\mu\text{m}$
Average kinetic flame density 2.68 kg/m^3
Average kinetic flame temperature $1,483.68 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $1.16\text{e}+07 \text{ W/m}^2$
Kinetic flame height $8.54 \text{ }\mu\text{m}$
Average kinetic flame density 3.62 kg/m^3
Average kinetic flame temperature $1,096.68 \text{ K}$
Heat flux density due to metal combustion, $7.54\text{e}+07 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $1.76\text{e}-06 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $9.09\text{e}-06 \text{ m}$
Pore diameter $1.93\text{e}-08 \text{ m}$
Diffusion flame heat flux density $1.32\text{e}+06 \text{ W/m}^2$
Diffusion flame height $231.98 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $3.29\text{e}+07 \text{ W/m}^2$

Heat flux density to surface outside skeletal structure $1.14\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $3.53\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $3.53\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0012 W/m^2

At pressure 4.06 MPa

Experimental burning rate 38.01 mm/s
Calculated burning rate 26.81 mm/s

First Conditional Fuel

Surface temperature 668.74 K
Linear burning rate 63.46 mm/s
Mass decomposition rate $105.27 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.18\text{e}+08 \text{ W/m}^2$
Kinetic flame height $1.39 \mu\text{m}$
Average kinetic flame density 7.36 kg/m^3
Average kinetic flame temperature 2,187.87 K
Binder sublimation heat flux density $2.18\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.035 W/m^2

Secondary Conditional Fuel

Surface temperature 605.43 K
Linear burning rate 11.82 mm/s
Mass decomposition rate $19.62 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $1.04\text{e}+07 \text{ W/m}^2$
Kinetic flame height $16.93 \mu\text{m}$
Average kinetic flame density 10.84 kg/m^3
Average kinetic flame temperature 1,485.21 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $6.37\text{e}+07 \text{ W/m}^2$
Kinetic flame height $1.55 \mu\text{m}$
Average kinetic flame density 14.66 kg/m^3
Average kinetic flame temperature 1,098.21 K
Heat flux density due to metal combustion, $8.24\text{e}+07 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity $0.4 \text{ W}/(\text{m}\cdot\text{K})$
Radiative thermal conductivity $1.49\text{e}-06 \text{ W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $8.3\text{e}-06 \text{ m}$
Pore diameter $1.61\text{e}-08 \text{ m}$
Diffusion flame heat flux density $1.2\text{e}+06 \text{ W/m}^2$
Diffusion flame height $253.99 \mu\text{m}$
Heat flux density to surface within skeletal structure $2.93\text{e}+07 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $8.31\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $3.88\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $3.88\text{e}+07 \text{ W/m}^2$
Surface heat balance error 0.0043 W/m^2

At pressure 6.5 MPa

Experimental burning rate 52.88 mm/s
Calculated burning rate 33.07 mm/s

First Conditional Fuel

Surface temperature 694.57 K
Linear burning rate 115.33 mm/s
Mass decomposition rate $191.33 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $4.04\text{e}+08 \text{ W/m}^2$
Kinetic flame height $0.75 \mu\text{m}$
Average kinetic flame density 11.72 kg/m^3
Average kinetic flame temperature 2,200.78 K
Binder sublimation heat flux density $4.04\text{e}+08 \text{ W/m}^2$

Surface heat balance error 0.0098 W/m²

Secondary Conditional Fuel

Surface temperature 611.78 K

Linear burning rate 14.22 mm/s

Mass decomposition rate 23.59 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.6e+07 W/m²

Kinetic flame height 10.93 μm

Average kinetic flame density 17.33 kg/m³

Average kinetic flame temperature 1,488.39 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1e+08 W/m²

Kinetic flame height 0.98 μm

Average kinetic flame density 23.42 kg/m³

Average kinetic flame temperature 1,101.39 K

Heat flux density due to metal combustion, 9.87e+07 W/m²

Effective thermal conductivity 0.4 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 1.06e-06 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 6.9e-06 m

Pore diameter 1.11e-08 m

Diffusion flame heat flux density 999,999.92 W/m²

Diffusion flame height 305.42 μm

Heat flux density to surface within skeletal structure 3.24e+07 W/m²

Heat flux density to surface outside skeletal structure 1.34e+07 W/m²

Total heat flux density to surface 4.69e+07 W/m²

Binder sublimation heat flux density 4.69e+07 W/m²

Surface heat balance error -0.0062 W/m²

Propellant Composition "Bas_0"

At pressure 1 MPa

Experimental burning rate 5.97 mm/s

Calculated burning rate 6.63 mm/s

First Conditional Fuel

Surface temperature 780 K

Linear burning rate 11.39 mm/s

Mass decomposition rate 18.89 kg·s⁻¹·m⁻²

Kinetic flame heat flux density to surface 4.23e+07 W/m²

Kinetic flame height 6.92 μm

Average kinetic flame density 1.77 kg/m³

Average kinetic flame temperature 2,243.5 K

Binder sublimation heat flux density 4.23e+07 W/m²

Surface heat balance error -0.0054 W/m²

Secondary Conditional Fuel

Surface temperature 725.78 K

Linear burning rate 3.02 mm/s

Mass decomposition rate 5 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 6.78e+06 W/m²

Kinetic flame height 24.19 μm

Average kinetic flame density 2.57 kg/m³

Average kinetic flame temperature 1,545.39 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 69,614.5 W/m²

Kinetic flame height 1,242.87 μm

Average kinetic flame density 3.43 kg/m³

Average kinetic flame temperature 1,158.39 K

Heat flux density due to metal combustion, $4.71\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $1.8\text{e}-05 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $1.35\text{e}-04 \text{ m}$
Pore diameter $1.14\text{e}-07 \text{ m}$
Diffusion flame heat flux density $4.54\text{e}+06 \text{ W/m}^2$
Diffusion flame height $64.79 \mu\text{m}$
Heat flux density to surface within skeletal structure $1.24\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $5.02\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $1.08\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $1.08\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0011 W/m^2

At pressure 4.06 MPa

Experimental burning rate 13.63 mm/s
Calculated burning rate 13.97 mm/s

First Conditional Fuel

Surface temperature 866.38 K
Linear burning rate 67.07 mm/s
Mass decomposition rate $111.27 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.64\text{e}+08 \text{ W/m}^2$
Kinetic flame height $1.08 \mu\text{m}$
Average kinetic flame density 7.04 kg/m^3
Average kinetic flame temperature $2,286.69 \text{ K}$
Binder sublimation heat flux density $2.64\text{e}+08 \text{ W/m}^2$
Surface heat balance error 0.031 W/m^2

Secondary Conditional Fuel

Surface temperature 752.33 K
Linear burning rate 5.92 mm/s
Mass decomposition rate $9.82 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $2.16\text{e}+07 \text{ W/m}^2$
Kinetic flame height $7.46 \mu\text{m}$
Average kinetic flame density 10.33 kg/m^3
Average kinetic flame temperature $1,558.67 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $217,325.01 \text{ W/m}^2$
Kinetic flame height $385.9 \mu\text{m}$
Average kinetic flame density 13.74 kg/m^3
Average kinetic flame temperature $1,171.67 \text{ K}$
Heat flux density due to metal combustion, $9.08\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $5.19\text{e}-06 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $6.88\text{e}-05 \text{ m}$
Pore diameter $2.96\text{e}-08 \text{ m}$
Diffusion flame heat flux density $2.29\text{e}+06 \text{ W/m}^2$
Diffusion flame height $127.21 \mu\text{m}$
Heat flux density to surface within skeletal structure $1.74\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $1.76\text{e}+07 \text{ W/m}^2$
Total heat flux density to surface $2.16\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $2.16\text{e}+07 \text{ W/m}^2$
Surface heat balance error 0.0023 W/m^2

At pressure 6.5 MPa

Experimental burning rate 18 mm/s

Calculated burning rate 18.99 mm/s

First Conditional Fuel

Surface temperature 900 K

Linear burning rate 121.98 mm/s

Mass decomposition rate $202.37 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$

Kinetic flame heat flux density to surface $4.9\text{e}+08 \text{ W/m}^2$

Kinetic flame height $0.57 \text{ } \mu\text{m}$

Average kinetic flame density 11.2 kg/m^3

Average kinetic flame temperature 2,303.5 K

Binder sublimation heat flux density $4.9\text{e}+08 \text{ W/m}^2$

Surface heat balance error 0.039 W/m^2

Secondary Conditional Fuel

Surface temperature 764.65 K

Linear burning rate 7.97 mm/s

Mass decomposition rate $13.22 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $2.98\text{e}+07 \text{ W/m}^2$

Kinetic flame height $5.37 \text{ } \mu\text{m}$

Average kinetic flame density 16.49 kg/m^3

Average kinetic flame temperature 1,564.83 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $304,713.2 \text{ W/m}^2$

Kinetic flame height $271.19 \text{ } \mu\text{m}$

Average kinetic flame density 21.9 kg/m^3

Average kinetic flame temperature 1,177.83 K

Heat flux density due to metal combustion, $1.21\text{e}+07 \text{ W/m}^2$

Effective thermal conductivity $0.4 \text{ W/(m} \cdot \text{K)}$

Conductive thermal conductivity $0.4 \text{ W/(m} \cdot \text{K)}$

Radiative thermal conductivity $3.01\text{e}-06 \text{ W/(m} \cdot \text{K)}$

Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$

Skeleton layer thickness $5.11\text{e}-05 \text{ m}$

Pore diameter $1.63\text{e}-08 \text{ m}$

Diffusion flame heat flux density $1.69\text{e}+06 \text{ W/m}^2$

Diffusion flame height $171.21 \text{ } \mu\text{m}$

Heat flux density to surface within skeletal structure $1.57\text{e}+06 \text{ W/m}^2$

Heat flux density to surface outside skeletal structure $2.6\text{e}+07 \text{ W/m}^2$

Total heat flux density to surface $2.93\text{e}+07 \text{ W/m}^2$

Binder sublimation heat flux density $2.93\text{e}+07 \text{ W/m}^2$

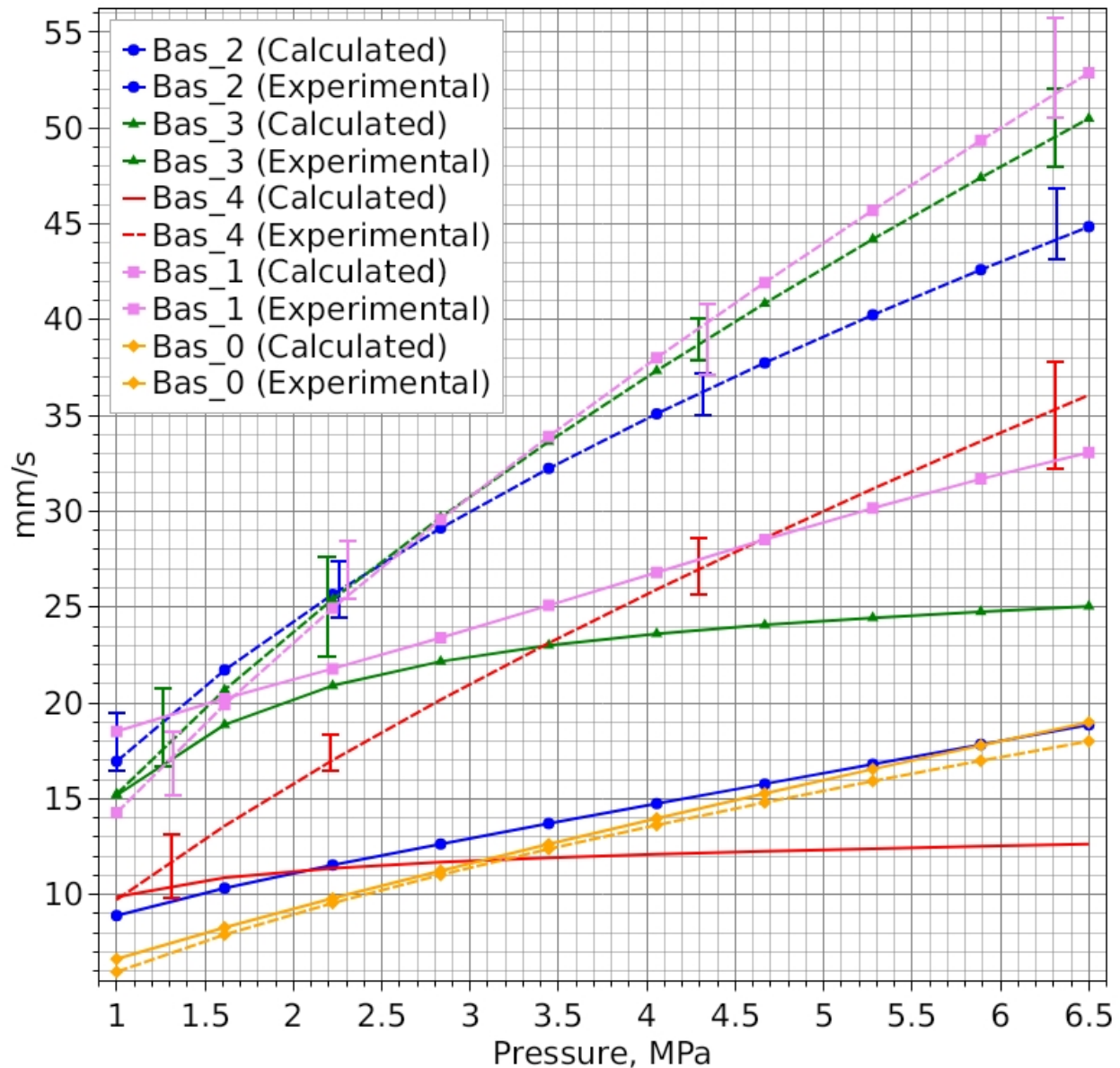
Surface heat balance error 0.0037 W/m^2

1 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	16.94 mm/s		15.24 mm/s		9.73 mm/s		14.26 mm/s		5.97 mm/s	
Calculated Burning Rate	8.9 mm/s		15.18 mm/s		9.89 mm/s		18.51 mm/s		6.63 mm/s	
Difference	8.04 mm/s		0.061 mm/s		-0.16 mm/s		-4.25 mm/s		-0.67 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	11.37 mm/s	4.21 mm/s	11.37 mm/s	11.37 mm/s	11.37 mm/s	3.37 mm/s	10.8 mm/s	10.8 mm/s	11.39 mm/s	3.02 mm/s
Surface Temperature	774.39 K	733.43 K	774.39 K	774.38 K	774.39 K	724.86 K	602.35 K	602.35 K	780 K	725.78 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	4.21e+07 W/m ²		4.21e+07 W/m ²		4.21e+07 W/m ²		3.53e+07 W/m ²		4.23e+07 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	2e+07 W/m ²		9.61e+07 W/m ²		5.33e+06 W/m ²		1.16e+07 W/m ²		69,614.5 W/m ²	
Kinetic Flame Height (Skeleton)	4.28 μm		1.92 μm		9.68 μm		8.54 μm		1,242.87 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	4.86e+06 W/m ²		1.82e+06 W/m ²		131,802.77 W/m ²		1.83e+06 W/m ²		6.78e+06 W/m ²	
Kinetic Flame Height (Out-Skeleton)	33.56 μm		87.63 μm		240.62 μm		96.55 μm		24.19 μm	
Diffusion Flame Heat Flux Density	3.17e+06 W/m ²		8.65e+06 W/m ²		2.22e+06 W/m ²		1.32e+06 W/m ²		4.54e+06 W/m ²	
Diffusion Flame Height	90.46 μm		33.51 μm		128.81 μm		231.98 μm		64.79 μm	
Metal Combustion Heat Flux Density	4.68e+06 W/m ²		6.77e+06 W/m ²		1.32e+07 W/m ²		7.54e+07 W/m ²		4.71e+06 W/m ²	

4.06 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	35.09 mm/s		37.34 mm/s		25.92 mm/s		38.01 mm/s		13.63 mm/s	
Calculated Burning Rate	14.74 mm/s		23.61 mm/s		12.1 mm/s		26.81 mm/s		13.97 mm/s	
Difference	20.35 mm/s		13.73 mm/s		13.82 mm/s		11.2 mm/s		-0.35 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	66.96 mm/s	6.26 mm/s	66.96 mm/s	11.37 mm/s	66.96 mm/s	3.48 mm/s	63.46 mm/s	11.82 mm/s	67.07 mm/s	5.92 mm/s
Surface Temperature	860.13 K	749.24 K	860.13 K	774.39 K	860.13 K	726.1 K	668.74 K	605.43 K	866.38 K	752.33 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	2.62e+08 W/m ²		2.62e+08 W/m ²		2.62e+08 W/m ²		2.18e+08 W/m ²		2.64e+08 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	1.42e+07 W/m ²		9.95e+07 W/m ²		5.35e+06 W/m ²		6.37e+07 W/m ²		217,325.01 W/m ²	
Kinetic Flame Height (Skeleton)	5.92 μm		1.86 μm		9.62 μm		1.55 μm		385.9 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	2.04e+07 W/m ²		1.13e+07 W/m ²		794,245.45 W/m ²		1.04e+07 W/m ²		2.16e+07 W/m ²	
Kinetic Flame Height (Out-Skeleton)	7.91 μm		14.08 μm		39.77 μm		16.93 μm		7.46 μm	
Diffusion Flame Heat Flux Density	2.12e+06 W/m ²		8.65e+06 W/m ²		2.14e+06 W/m ²		1.2e+06 W/m ²		2.29e+06 W/m ²	
Diffusion Flame Height	134.48 μm		33.51 μm		133.05 μm		253.99 μm		127.21 μm	
Metal Combustion Heat Flux Density	6.89e+06 W/m ²		6.77e+06 W/m ²		1.36e+07 W/m ²		8.24e+07 W/m ²		9.08e+06 W/m ²	

6.5 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	44.85 mm/s		50.5 mm/s		36.06 mm/s		52.88 mm/s		18 mm/s	
Calculated Burning Rate	18.86 mm/s		25.04 mm/s		12.63 mm/s		33.07 mm/s		18.99 mm/s	
Difference	25.99 mm/s		25.46 mm/s		23.43 mm/s		19.81 mm/s		-0.99 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	121.79 mm/s	7.91 mm/s	121.79 mm/s	11.45 mm/s	121.79 mm/s	3.58 mm/s	115.33 mm/s	14.22 mm/s	121.98 mm/s	7.97 mm/s
Surface Temperature	893.51 K	758.92 K	893.51 K	774.69 K	893.51 K	727.1 K	694.57 K	611.78 K	900 K	764.65 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	4.87e+08 W/m ²		4.87e+08 W/m ²		4.87e+08 W/m ²		4.04e+08 W/m ²		4.9e+08 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	1.15e+07 W/m ²		1e+08 W/m ²		5.29e+06 W/m ²		1e+08 W/m ²		304,713.2 W/m ²	
Kinetic Flame Height (Skeleton)	7.23 μm		1.85 μm		9.72 μm		0.98 μm		271.19 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	3e+07 W/m ²		2.08e+07 W/m ²		1.43e+06 W/m ²		1.6e+07 W/m ²		2.98e+07 W/m ²	
Kinetic Flame Height (Out-Skeleton)	5.36 μm		7.65 μm		21.98 μm		10.93 μm		5.37 μm	
Diffusion Flame Heat Flux Density	1.67e+06 W/m ²		8.59e+06 W/m ²		2.09e+06 W/m ²		999,999.92 W/m ²		1.69e+06 W/m ²	
Diffusion Flame Height	170.01 μm		33.74 μm		136.57 μm		305.42 μm		171.21 μm	
Metal Combustion Heat Flux Density	8.66e+06 W/m ²		6.81e+06 W/m ²		1.4e+07 W/m ²		9.87e+07 W/m ²		1.21e+07 W/m ²	

Burning Rates - Group Optimization (All Propellants)



Propellant Composition Parameters

Composition "Bas_2"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 1.285341E-05$; $v = 0.52$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
 Average molar mass 0.033 kg/mol
 Specific heat capacity (constant volume) 635.6 J/kg.K
 Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
 Average molar mass 0.033 kg/mol
 Specific heat capacity (constant volume) 635.6 J/kg.K
 Diffusion flame temperature 3,604 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
 Average molar mass 0.033 kg/mol
 Specific heat capacity (constant volume) 635.6 J/kg.K
 Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_3"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.57.

Vieille's Power Law: $U = A \cdot p^v$, $A = 2.203002E-06$; $v = 0.64$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,674 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,623 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_4"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.56.

Vieille's Power Law: $U = A \cdot p^v$, $A = 6.137894E-07$; $v = 0.7$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,580 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,241 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,042 K

Composition "Bas_1"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 9.000448E-07$; $v = 0.7$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 2,365 K

Composition "Bas_0"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 1.720582E-06$; $v = 0.59$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

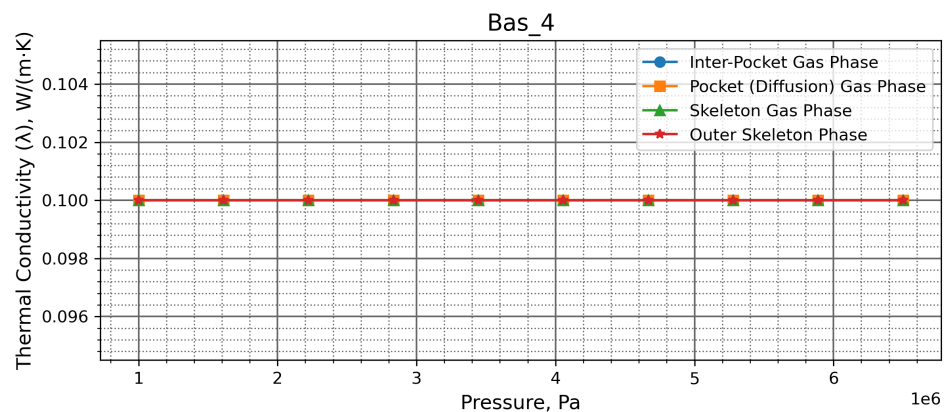
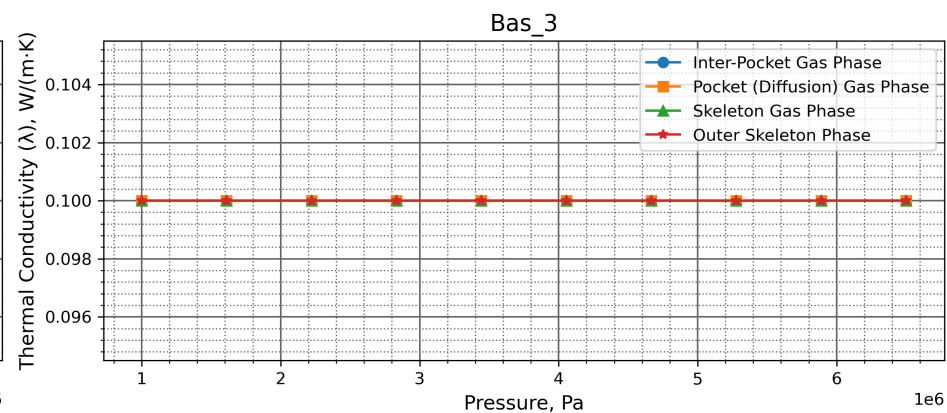
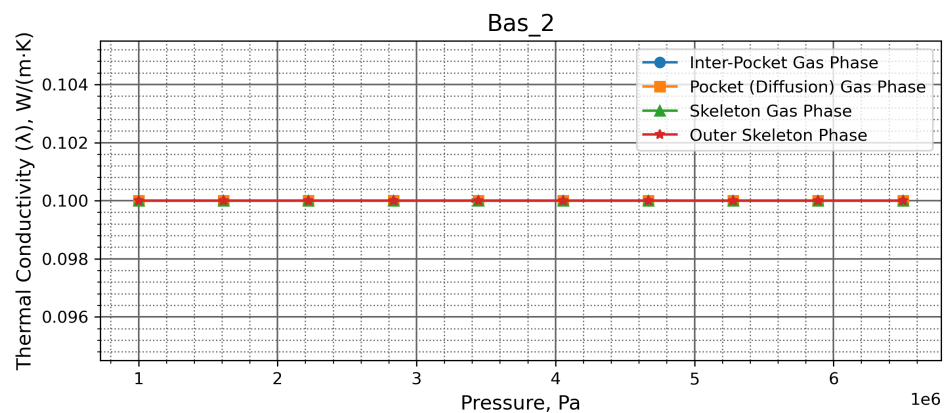
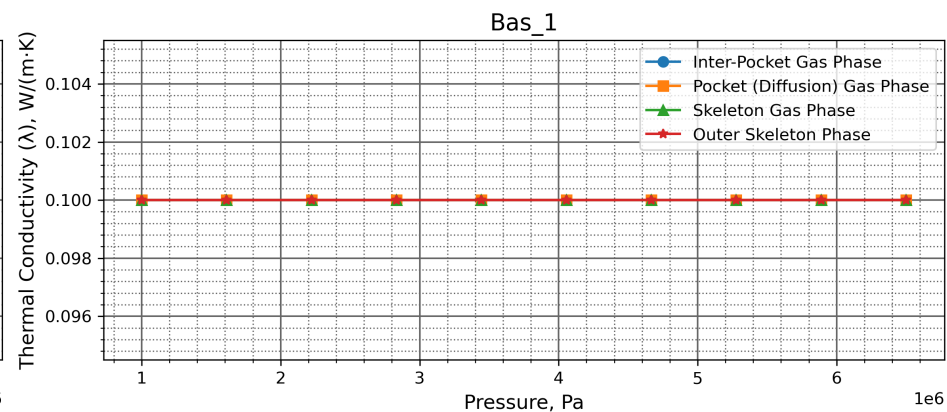
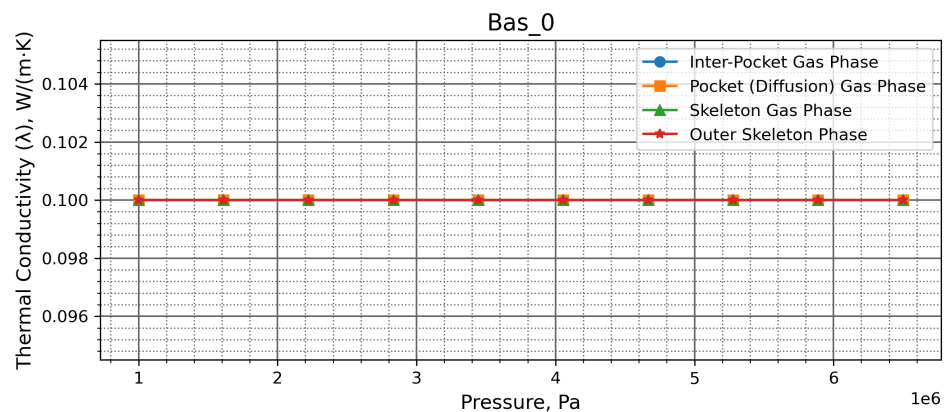
Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

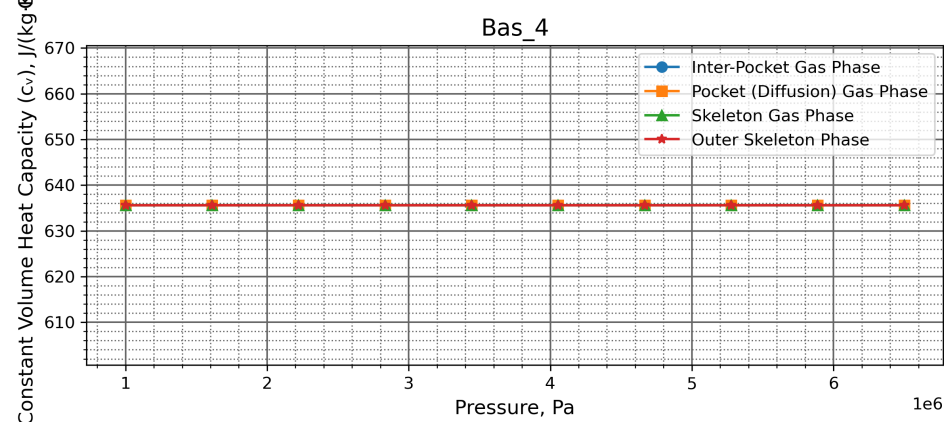
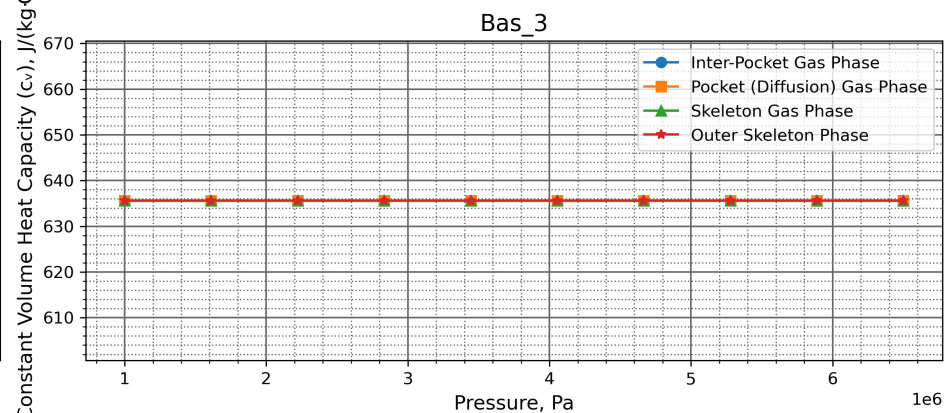
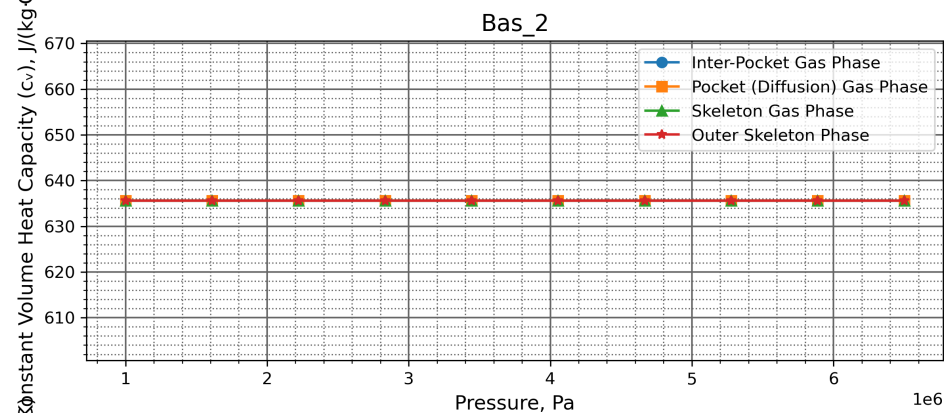
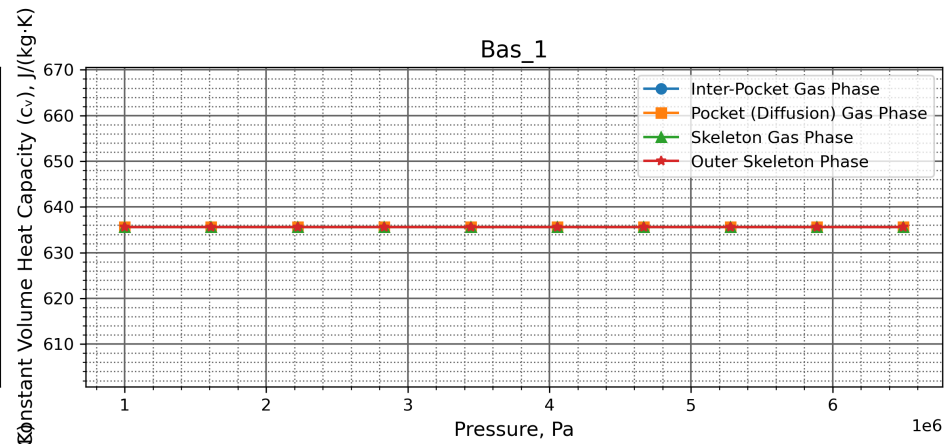
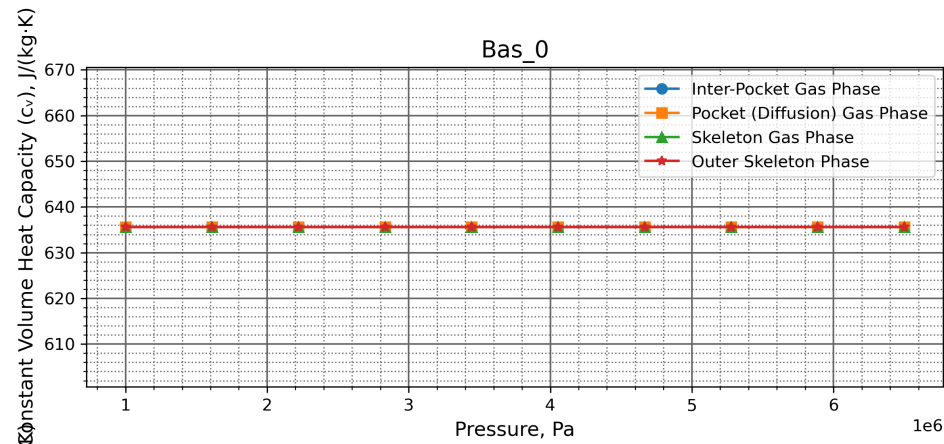
Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 2,365 K

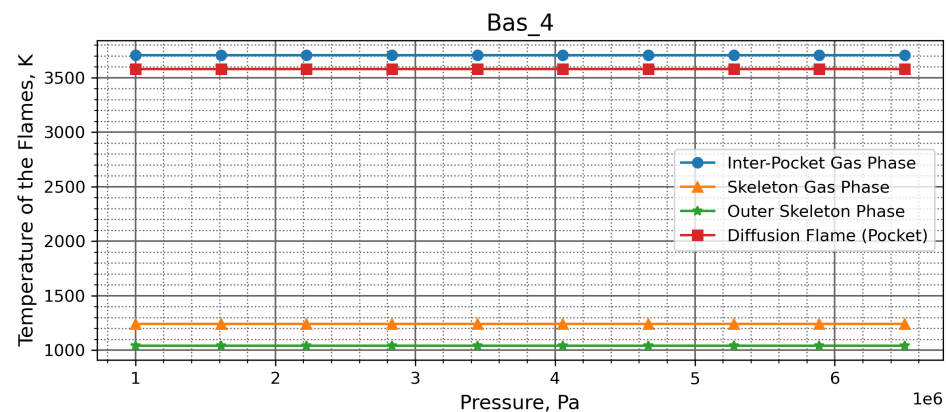
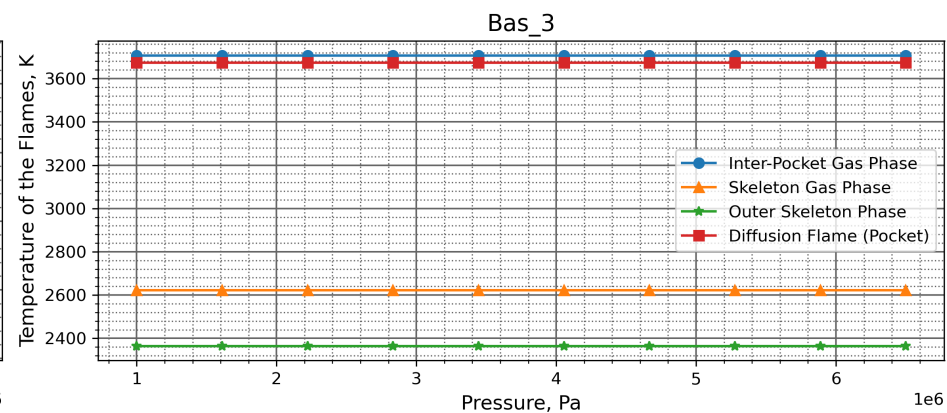
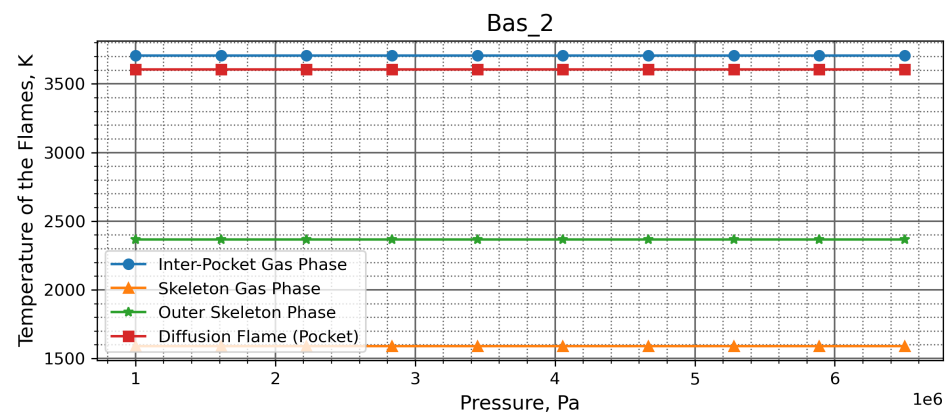
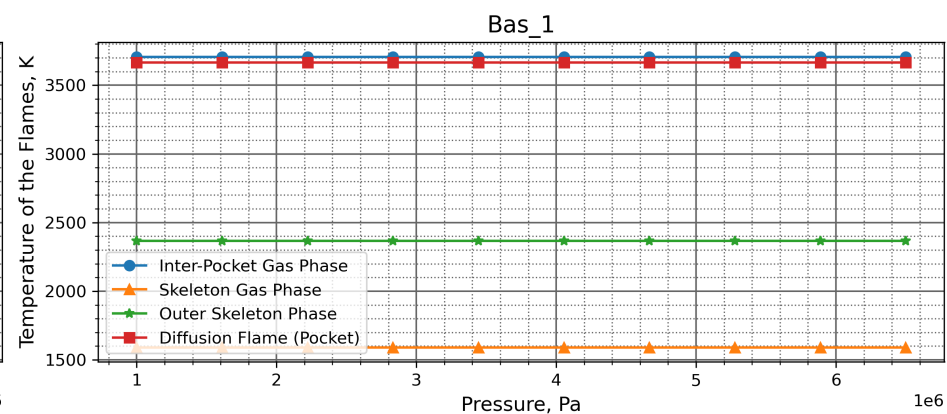
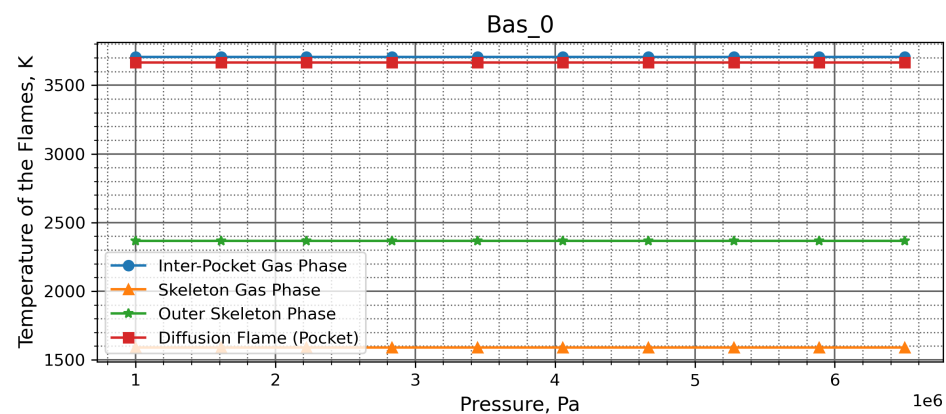
Thermal Conductivity (λ), W/(m·K)



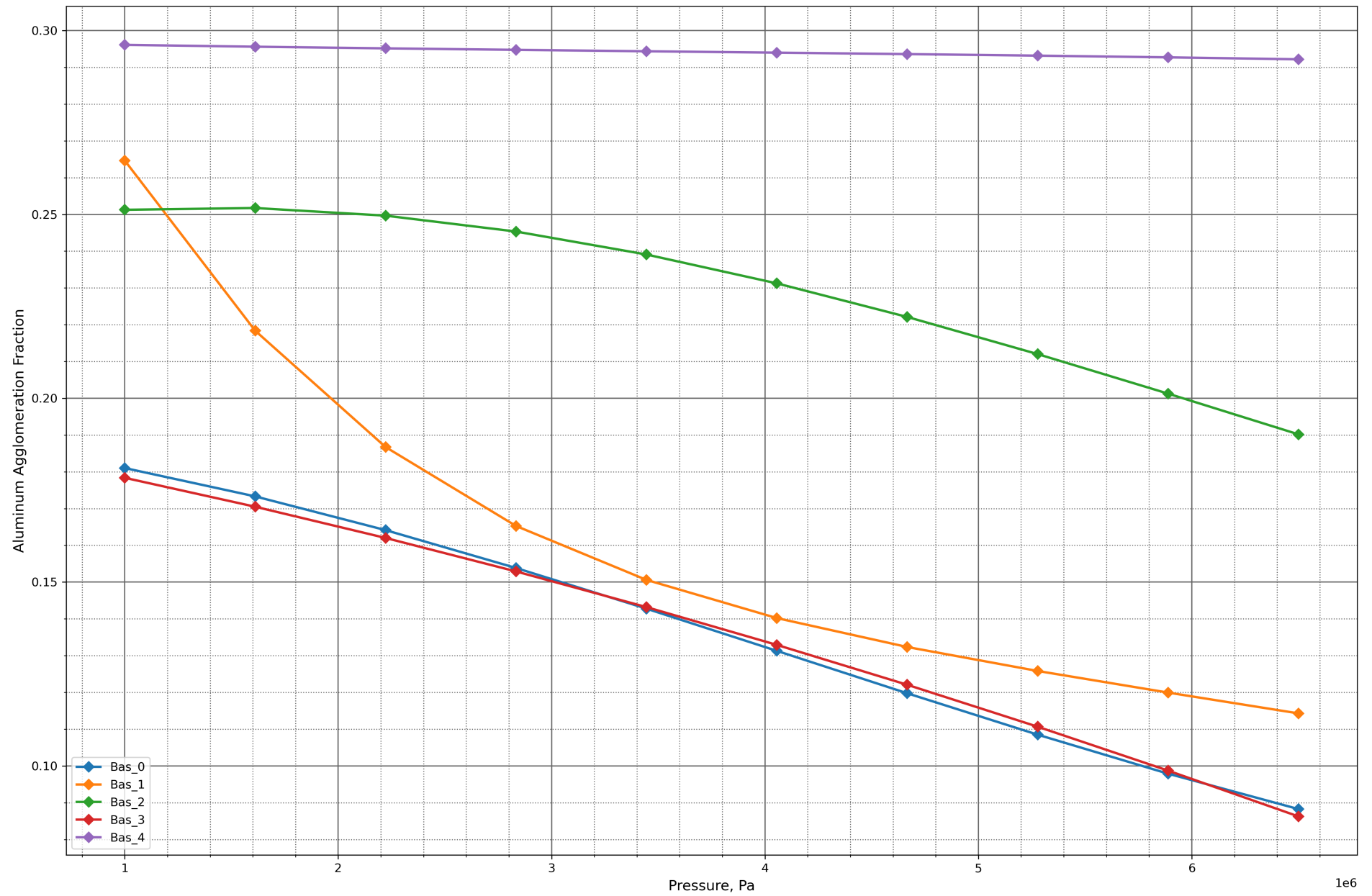
Constant Volume Heat Capacity (c_v), J/(kg·K)



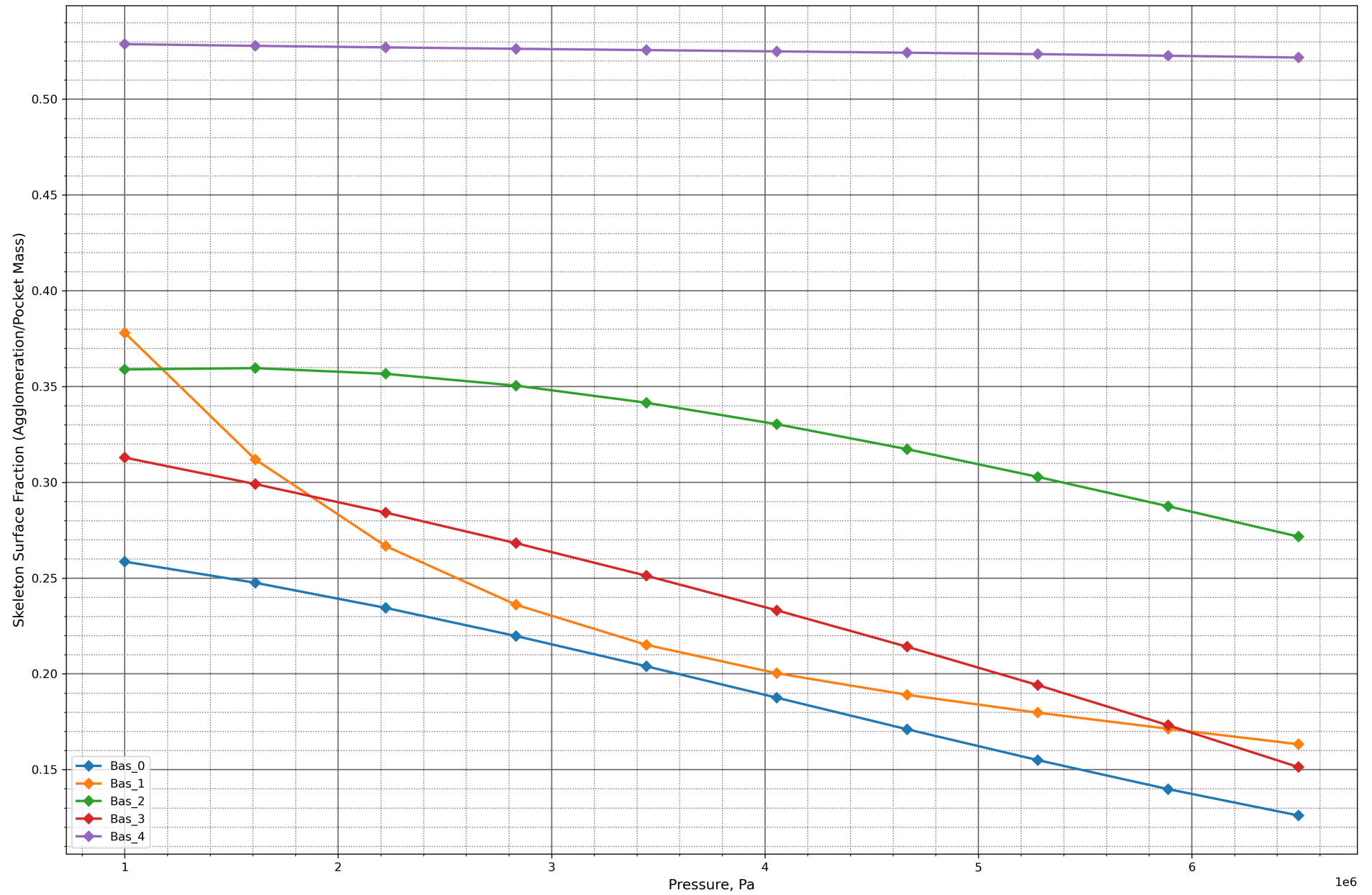
Temperature of the Flames, K



Aluminum Agglomeration Fraction



Skeleton Surface Fraction (Agglomeration/Pocket Mass)



Performance Report

System Information

Operating System: Unix 6.8.0.117 (Ubuntu 24.04.4 LTS)
Processor: Intel(R) Xeon(R) CPU E5-2697 v4 @ 2.30GHz (Base Clock Frequency 2.29 GHz)
Memory: 15.6 GB (64-bit), Unknown
Machine Name: vscode
Runtime: .NET 10.0.8 (.NET 10.0.8)
Architecture: X64 / X64

Total Execution Time

Total execution time: 02:22:07.643

Garbage Collection Information

Total Memory: 7037624 bytes
GC Pause Time Percentage: 0.14%
Generation 0 Collections: 10012
Generation 1 Collections: 849
Generation 2 Collections: 85

Detailed Performance Metrics

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 5939940

Maximum execution time: 30.4781 ms

Minimum execution time: 0.0001 ms

Mean execution time: 0.8340913174208143 ms

Execution time standard deviation: 0.2181088154788975 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 5532808

Maximum execution time: 13.7868 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.1996849079527113 ms

Execution time standard deviation: 0.13289327937390863 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 3609380

Maximum execution time: 12.8194 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2780822255345456 ms

Execution time standard deviation: 0.06162051164900342 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 5939556

Maximum execution time: 10.0649 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.8504653411467287 ms

Execution time standard deviation: 0.2085737967179505 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 5534119

Maximum execution time: 8.6152 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.20319062018724435 ms

Execution time standard deviation: 0.13332286354743011 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 3612471

Maximum execution time: 8.3244 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.2820467602369565 ms
Execution time standard deviation: 0.05602995041027123 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 5939556
Maximum execution time: 15.2003 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8344893322665795 ms
Execution time standard deviation: 0.2166494031807024 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 5533328
Maximum execution time: 19.3398 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.19983724639133257 ms
Execution time standard deviation: 0.1332337986597693 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 3608481
Maximum execution time: 6.5464 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2769181581945405 ms
Execution time standard deviation: 0.060012826255010775 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 5939556
Maximum execution time: 23.1468 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8344448675287917 ms
Execution time standard deviation: 0.2171274877760539 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 5533195
Maximum execution time: 10.304 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.1992448731700277 ms
Execution time standard deviation: 0.13249793063548118 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 3608207
Maximum execution time: 13.556 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.27707809698835806 ms
Execution time standard deviation: 0.06065340630626449 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 5939556
Maximum execution time: 13.0797 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8226186043536924 ms
Execution time standard deviation: 0.22141519407584664 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 5532363
Maximum execution time: 9.925 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.1965032904927 ms
Execution time standard deviation: 0.13145582013441684 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance

Call count: 3608734

Maximum execution time: 11.9191 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.27272305013335285 ms

Execution time standard deviation: 0.06261081361069915 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 5939556

Maximum execution time: 20.3382 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.838451844666521 ms

Execution time standard deviation: 0.21516557639488154 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 5532456

Maximum execution time: 7.4847 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2002923550950904 ms

Execution time standard deviation: 0.1324952810584401 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 3610337

Maximum execution time: 10.9657 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2784132357450312 ms

Execution time standard deviation: 0.05893908590776908 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 5939556

Maximum execution time: 23.2483 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.8417579584905761 ms

Execution time standard deviation: 0.21465715509910277 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 5532826

Maximum execution time: 6.734 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.20097160653525983 ms

Execution time standard deviation: 0.13278047201362936 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 3610236

Maximum execution time: 8.3632 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.27912558577886903 ms

Execution time standard deviation: 0.05856722072299315 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 5939556

Maximum execution time: 12.1279 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.8359317203171799 ms

Execution time standard deviation: 0.215581821895583 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 5533542

Maximum execution time: 17.9642 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.19992187725330732 ms
Execution time standard deviation: 0.13286744240772208 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 3612398
Maximum execution time: 20.5081 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2777212464961058 ms
Execution time standard deviation: 0.06083243339765304 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 5939556
Maximum execution time: 14.9244 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8340721145149641 ms
Execution time standard deviation: 0.21754743032243035 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 5532257
Maximum execution time: 11.5136 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.19943743873431682 ms
Execution time standard deviation: 0.13260723820038844 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 3610251
Maximum execution time: 14.98 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.27687229685693815 ms
Execution time standard deviation: 0.06041624834825 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 5939556
Maximum execution time: 18.0174 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8241723368885129 ms
Execution time standard deviation: 0.22127168769826153 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 5533180
Maximum execution time: 17.6522 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.19693977510219512 ms
Execution time standard deviation: 0.13205844121330862 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 3607598
Maximum execution time: 9.288 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.27436708103285673 ms
Execution time standard deviation: 0.061943874687666364 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 5939556
Maximum execution time: 16.1597 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8390493902238179 ms

Execution time standard deviation: 0.2147900210654809 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 5532115

Maximum execution time: 9.279 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2006775193574204 ms

Execution time standard deviation: 0.13263206979023892 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 3611215

Maximum execution time: 12.4537 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2787650695680052 ms

Execution time standard deviation: 0.058514622389929084 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 5939556

Maximum execution time: 27.2658 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.8341442805151937 ms

Execution time standard deviation: 0.21782243760776243 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 5531793

Maximum execution time: 14.6962 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.19994214085739903 ms

Execution time standard deviation: 0.13278653800956297 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 3610827

Maximum execution time: 14.9248 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.27680303412486174 ms

Execution time standard deviation: 0.060065553021529265 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 5939556

Maximum execution time: 35.4928 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.8365686407031145 ms

Execution time standard deviation: 0.21636714037439606 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 5532040

Maximum execution time: 4.3611 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.20002590109616877 ms

Execution time standard deviation: 0.1323270465285004 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 3609996

Maximum execution time: 15.1655 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2781253260668195 ms

Execution time standard deviation: 0.059631586322007685 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 5939556
Maximum execution time: 33.8731 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8252799074711755 ms
Execution time standard deviation: 0.2206258186933723 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 5532629
Maximum execution time: 9.0614 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.19695524892414015 ms
Execution time standard deviation: 0.13147929573090197 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 3609554
Maximum execution time: 8.1014 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.27385472360299645 ms
Execution time standard deviation: 0.06051785729220461 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 5939556
Maximum execution time: 16.9731 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8329092498328247 ms
Execution time standard deviation: 0.21740031870625814 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 5534030
Maximum execution time: 11.6904 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.1992152437554439 ms
Execution time standard deviation: 0.13250246825380815 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 3612350
Maximum execution time: 21.0561 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.27623572040358835 ms
Execution time standard deviation: 0.06163235072165264 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 5939556
Maximum execution time: 19.05 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8365929855194488 ms
Execution time standard deviation: 0.21624653121256238 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 5532333
Maximum execution time: 13.876 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.19972237007425375 ms
Execution time standard deviation: 0.13235342039756917 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 3610650
Maximum execution time: 6.3432 ms
Minimum execution time: 0.0002 ms

Mean execution time: 0.2775282053092878 ms
Execution time standard deviation: 0.05870536561122456 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance

Call count: 5939556
Maximum execution time: 18.4448 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8310501319458816 ms
Execution time standard deviation: 0.21866485413893755 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance

Call count: 5531454
Maximum execution time: 7.5317 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.1986564853472532 ms
Execution time standard deviation: 0.13210973308131266 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance

Call count: 3608906
Maximum execution time: 5.9041 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.27572791416566456 ms
Execution time standard deviation: 0.05960042167247148 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance

Call count: 5939556
Maximum execution time: 18.8243 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.824925801474039 ms
Execution time standard deviation: 0.22111389297932166 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance

Call count: 5531801
Maximum execution time: 10.5264 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.1970291362975921 ms
Execution time standard deviation: 0.1314119503384138 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance

Call count: 3608935
Maximum execution time: 12.2871 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.27398610847797616 ms
Execution time standard deviation: 0.0612670322519454 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance

Call count: 5939556
Maximum execution time: 23.8492 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8383365068028882 ms
Execution time standard deviation: 0.21482753309622205 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance

Call count: 5532686
Maximum execution time: 14.0364 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.20044667064061414 ms
Execution time standard deviation: 0.1327400760071969 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 3609442
Maximum execution time: 12.4181 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2789515490205996 ms
Execution time standard deviation: 0.05888847427491997 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 5939556
Maximum execution time: 16.6926 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8370609131558256 ms
Execution time standard deviation: 0.21523201214530788 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 5532200
Maximum execution time: 9.1056 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.20002646943348665 ms
Execution time standard deviation: 0.13231009297800575 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 3610357
Maximum execution time: 15.4339 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.27790356895454515 ms
Execution time standard deviation: 0.05873232232068562 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 5939556
Maximum execution time: 15.824 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8335638534934103 ms
Execution time standard deviation: 0.21606717217924448 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 5532801
Maximum execution time: 10.0747 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.19952496858281718 ms
Execution time standard deviation: 0.1322731808621624 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 3608119
Maximum execution time: 9.3568 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2766806733369656 ms
Execution time standard deviation: 0.0589965835054036 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 5939556
Maximum execution time: 17.4086 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8457793549383119 ms
Execution time standard deviation: 0.21076071919716352 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 5533604
Maximum execution time: 6.4444 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.20233492505063938 ms
Execution time standard deviation: 0.132746950975429 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 3612181
Maximum execution time: 5.2044 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2804689333120383 ms
Execution time standard deviation: 0.05568078375691401 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 5939556
Maximum execution time: 25.964 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8265012249063296 ms
Execution time standard deviation: 0.21983253298126337 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 5532399
Maximum execution time: 14.998 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.19768820645437404 ms
Execution time standard deviation: 0.13170944529315198 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 3609757
Maximum execution time: 15.0671 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2740243820013247 ms
Execution time standard deviation: 0.06093765847218476 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 5939556
Maximum execution time: 19.9346 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8256435331192721 ms
Execution time standard deviation: 0.21962443496343806 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 5533299
Maximum execution time: 11.5881 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.1974370112657917 ms
Execution time standard deviation: 0.13150174437443646 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 3610659
Maximum execution time: 16.1186 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2740043091025808 ms
Execution time standard deviation: 0.06116423075034969 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance
Call count: 5939556
Maximum execution time: 18.2378 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8425991400872206 ms
Execution time standard deviation: 0.21344885346910736 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance

Call count: 5533027

Maximum execution time: 18.0039 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.20116267446372352 ms

Execution time standard deviation: 0.1329931323130191 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance

Call count: 3609272

Maximum execution time: 12.6606 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2795074350173413 ms

Execution time standard deviation: 0.05724218763093219 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 5939556

Maximum execution time: 20.3426 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.8397335856249187 ms

Execution time standard deviation: 0.2142499263383787 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 5534021

Maximum execution time: 13.5819 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.20068424362682727 ms

Execution time standard deviation: 0.1327329850005347 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 3613474

Maximum execution time: 10.3153 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2784603112129498 ms

Execution time standard deviation: 0.05779559533360995 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance

Call count: 5939556

Maximum execution time: 16.1929 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.8293782666078355 ms

Execution time standard deviation: 0.21803719543560315 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance

Call count: 5533296

Maximum execution time: 7.0091 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.19804159217218598 ms

Execution time standard deviation: 0.1314533516811528 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance

Call count: 3610593

Maximum execution time: 28.0422 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2749804398612898 ms

Execution time standard deviation: 0.06199388745055163 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance

Call count: 5939556

Maximum execution time: 24.8507 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.84000423514826 ms
Execution time standard deviation: 0.21476335584959566 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance

Call count: 5532822
Maximum execution time: 11.3709 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2009335287851455 ms
Execution time standard deviation: 0.13269032019807758 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance

Call count: 3611237
Maximum execution time: 13.9423 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2793763088659155 ms
Execution time standard deviation: 0.05925595947903042 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance

Call count: 5939556
Maximum execution time: 17.7626 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8213972961278794 ms
Execution time standard deviation: 0.22024165881458801 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance

Call count: 5532274
Maximum execution time: 13.5757 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.19648966378023183 ms
Execution time standard deviation: 0.13119576813307993 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance

Call count: 3612709
Maximum execution time: 20.3438 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.27263419943316913 ms
Execution time standard deviation: 0.06240982980463045 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance

Call count: 5939556
Maximum execution time: 21.0049 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8410597430683403 ms
Execution time standard deviation: 0.21320741206147287 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance

Call count: 5532453
Maximum execution time: 7.0663 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.20147837460166254 ms
Execution time standard deviation: 0.13248106243455793 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance

Call count: 3609643
Maximum execution time: 6.8403 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.27987700409708877 ms

Execution time standard deviation: 0.05740143457050266 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance

Call count: 5939556

Maximum execution time: 25.7143 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.8494646097284181 ms

Execution time standard deviation: 0.21026261055969098 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance

Call count: 5531845

Maximum execution time: 6.6548 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.20301477915597443 ms

Execution time standard deviation: 0.13281731784538822 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance

Call count: 3611921

Maximum execution time: 10.3122 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.28196863890433627 ms

Execution time standard deviation: 0.05531077551376665 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance

Call count: 5939556

Maximum execution time: 16.9634 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.8380638542341856 ms

Execution time standard deviation: 0.2148971723312006 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance

Call count: 5533338

Maximum execution time: 9.3606 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2004551298872587 ms

Execution time standard deviation: 0.13242523816139676 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance

Call count: 3610345

Maximum execution time: 16.2908 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2779078731533946 ms

Execution time standard deviation: 0.05889486063219546 ms